

Learning to share sunlight: differentiable-simulation design and control of climate-resilient agrivoltaics across the world's staple croplands

Quang-Vinh Dang*

British University Vietnam, Hung Yen, Vietnam

Abstract

Solar power and food production increasingly compete for the same land, while warming exposes crops to more heat and drought. Agrivoltaic systems, which grow crops under photovoltaic (PV) modules, can relieve this conflict, but how much electricity farmland can produce without compromising food security depends on how sunlight is shared between modules and plants, day by day, for every crop and climate. We propose SALS (stress-aware light sharing), an algorithm that learns the design and the daily control of single-axis tracker agrivoltaics end-to-end through a differentiable simulator. The simulator couples solar geometry, backtracking trackers, PV output, crop-level light and microclimate, a soil-water balance and the SIMPLE crop model with heat and drought stress. A design network maps site descriptors to the ground coverage ratio, and a control network maps the daily crop, soil and weather state to the tracker's light-sharing level. Both are trained by back-propagation through whole seasons to maximise electricity subject to a 90% yield-retention floor. We applied SALS to 400 sites representing the global harvested area of wheat, rice, maize, soybean and potato, with NASA POWER weather for 2001–2019 and warming of +1.5, +2 and +3 °C. On geographically held-out sites and years, SALS produced 1019 MWh ha⁻¹ yr⁻¹ (72% of a conventional PV plant) at a mean yield retention of 0.91. This gave a land-equivalent ratio of 1.63, compared with 1.20 for static agrivoltaic designs and 1.42 for an expert stress rule ($p < 10^{-4}$). Its advantage grew with warming, to an LER of 1.75 at +3 °C, as the learned policy shifted from sharing light during canopy development to shading crops during heat. The yield floor was met on average but in only 45–49% of individual seasons, which calls for chance-constrained extensions. Scaled to 5% of the harvested

*Corresponding author.
Preprint submitted to Artificial Intelligence in Agriculture
Email address: vinh.dq4@buv.edu.vn (Quang-Vinh Dang)

area of the five crops, learned agrivoltaics has a technical potential of about 31,000 TWh yr⁻¹, comparable to current global electricity generation.

7 *Keywords:* Agrivoltaics, Differentiable simulation, Food–energy nexus,
8 Climate adaptation, Solar tracking, Crop modelling, Physics-informed
9 machine learning

10 **1. Introduction**

11 Two of the defining challenges of this century compete for the same re-
12 source: land. The energy transition requires terawatts of new solar pho-
13 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
14 to two hectares per megawatt. Food production must rise while crops are
15 exposed to more frequent heat waves and droughts. Warming alone is pro-
16 jected to reduce global yields of maize, wheat, rice and soybean by several
17 per cent per degree (Zhao et al., 2017), and a new generation of climate and
18 crop models shows these impacts emerging earlier than previously expected
19 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
20 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
21 rural electricity demand, so converting farmland into solar farms is tempting
22 and controversial at the same time.

23 Agrivoltaic (AV) systems combine crops and PV modules on the same
24 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
25 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
26 a hectare can deliver more food plus energy than the two uses side by side,
27 with land-equivalent ratios (LER) well above one in field trials and mod-
28 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
29 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
30 climate. It lowers canopy temperature and evaporative demand, conserves
31 soil moisture and improves water-use efficiency, especially in drylands (Mar-
32 rou et al., 2013; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
33 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
34 date adaptation measure against heat and drought stress, in addition to a
35 mitigation measure through renewable electricity. The same shade, however,
36 also reduces the light available for photosynthesis. Regulators increasingly
37 require AV systems to retain most of the reference crop yield. The German
38 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
39 reference yield (DIN, 2021), and other schemes and many farmers set stricter

40 floors. How much electricity can be produced without sacrificing food secu-
41 rity depends on the crop, the local climate, the year and the growth stage.

42 Most AV designs fix the geometry of the array, i.e. the ground coverage
43 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
44 generic rules of thumb. Single-axis trackers add a degree of freedom: the
45 modules can be rotated away from the sun to let light through when the crop
46 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
47 sun to maximise electricity and shade the crop during heat waves. Choosing
48 the rotation every day is a sequential decision problem. The value of light
49 depends on the crop’s phenological stage, soil water and upcoming weather,
50 and the effect of a decision appears in the harvest months later. Designing
51 the GCR and the control policy for every combination of crop and climate
52 on Earth, and for a warming climate, is beyond manual rule design. Recent
53 work in this journal has used evolutionary multi-objective optimisation to
54 lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for
55 crop management has mainly used reinforcement learning with black-box
56 simulators (Gautron et al., 2022), which is sample-inefficient and hard to
57 scale to global, multi-decadal assessments.

58 This paper proposes *SALS* (stress-aware light sharing), a learning algo-
59 rithm that designs and controls agrivoltaic systems jointly, for any cropland
60 site on Earth, through a differentiable physics simulator. The simulator
61 couples hourly solar geometry, single-axis trackers with backtracking, row-
62 to-row shading and PV electricity, the direct and diffuse light reaching the
63 crop, a microclimate model of canopy temperature and reference evapotran-
64 spiration under the modules, and the SIMPLE crop model with heat and
65 drought stress (Zhao et al., 2019). Because every component is written as
66 differentiable tensor operations, two small neural networks can be trained
67 by gradient descent through entire growing seasons. A *design network* maps
68 site descriptors to the GCR, and a *control network* maps the daily crop and
69 soil state and the same-day weather forecast to the tracker’s light-sharing
70 level. Training maximises electricity subject to a yield-retention floor, across
71 hundreds of sites, twenty years of weather and four warming levels. This
72 physics-informed learning approach (Karniadakis et al., 2021) gives a single
73 policy that generalises to unseen regions, years and climates.

74 We apply SALS to 400 sites that represent where maize, wheat, rice,
75 soybean and potato are grown worldwide, sampled in proportion to their
76 harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al.,
77 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks,

78 2018). The contributions are as follows.

- 79 1. A differentiable agrivoltaic simulator that couples tracker geometry, PV
80 output, crop-level light and microclimate, soil water and a generic crop
81 model, and runs thousands of site-seasons in parallel on a CPU.
- 82 2. The SALS algorithm, which learns the array design and a closed-loop,
83 stress-aware tracker policy end-to-end under a food-security constraint.
84 It is evaluated on geographically held-out sites and years against static
85 designs, expert rules and a perfect-foresight oracle.
- 86 3. A global assessment under warming of $+1.5$, $+2$ and $+3$ °C, which
87 quantifies how much electricity croplands can supply while retaining at
88 least 90% of crop yields, and where agrivoltaics protects yields against
89 heat and drought.

90 2. Materials and methods

91 2.1. Global cropland sites, weather and crop calendars

92 We studied five staple crops that together cover about 670 Mha of har-
93 vested area: wheat, rice, maize, soybean and potato. Harvested areas (rain-
94 fed and irrigated) were taken from the SPAM 2010 gridded production maps
95 (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with
96 at least 5000 ha of the crop were drawn without replacement with probabil-
97 ity proportional to harvested area. An unweighted average over the sample
98 therefore approximates an area-weighted average over global production. The
99 400 crop-site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing
100 and maturity dates came from the GGCM Phase 3 crop calendar at 0.5°
101 (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar un-
102 less the coldest-month mean temperature was below -8°C , in which case
103 the spring-wheat calendar was used. Daily weather for 2001–2020 was re-
104 trieved from NASA POWER (Sparks, 2018): global and diffuse shortwave
105 irradiance, maximum, minimum and dew-point temperature, precipitation
106 and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–
107 2019), each simulated as the 365 days after sowing, which gives 7600 site-
108 seasons.

109 Climate change was represented by the global warming levels $+1.5$, $+2$
110 and $+3$ °C relative to 2001–2020. Daily maximum, minimum and dew-point
111 temperatures were shifted uniformly, which keeps relative humidity approx-
112 imately constant and raises the vapour-pressure deficit. The atmospheric

CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.3.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

Hourly irradiance. Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

Array geometry and PV electricity. We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, which leaves space for machinery under the modules. The rotation ϕ is limited to $\pm 60^\circ$, and the ground coverage ratio g is the module width divided by the row pitch. Let ψ be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al., 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ otherwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines beam irradiance on the module (reduced by the fraction of the module shaded by the neighbouring row, $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$), isotropic sky diffuse and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature losses and system losses. A conventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the energy reference E_{PV} .

147 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 148 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 149 fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-
 150 view factor. We pre-computed it for 25 values of g and 13 rotation angles by
 151 ray casting over 31 neighbouring rows, and interpolate it bilinearly (differenti-
 152 ably). The daily ratio s of the irradiance at the crop to that of the open field
 153 defines the crop’s shading. Following observations of cooler canopies under
 154 PV modules (Marrou et al., 2013; Barron-Gafford et al., 2019), the canopy
 155 maximum temperature is lowered by $\kappa(1 - s)$ with $\kappa = 2^\circ\text{C}$ (a sensitivity
 156 analysis over κ is reported in Section 3.5). The reference evapotranspiration
 157 under the modules, ET_0 , is recomputed with the FAO-56 Penman–Monteith
 158 equation (Allen et al., 1998) from the reduced radiation and canopy temper-
 159 ature. Soil water conservation under the modules (Hassanpour Adeg et al.,
 160 2018; Elamri et al., 2018) therefore emerges from the model and is not im-
 161 posed.

162 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 163 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 164 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

165 where $R_{\text{crop}} = s R$ is the radiation received by the crop. f_{Solar} is the canopy’s
 166 radiation interception, driven by thermal time and accelerated senescence
 167 under heat and drought stress. f_{Heat} decreases linearly between the heat
 168 threshold T_{heat} and the extreme temperature T_{ext} of the canopy maximum
 169 temperature. $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, with the agricultural reference index
 170 for drought (Woli et al., 2012) computed from a daily soil-water balance with
 171 curve-number runoff, drainage and root water uptake. Irrigated fields have
 172 $f_{\text{Water}} = 1$. Yields of the rainfed and irrigated parts of a site are combined
 173 with the SPAM irrigated fraction. The thermal-time requirement to maturity
 174 was calibrated for every site as the mean thermal time between the GGCM
 175 sowing and maturity dates under the 2001–2020 climate, and the canopy
 176 parameters were scaled accordingly. The cultivar is thus adapted to the
 177 present climate, and warming shortens the season. Growth stops smoothly
 178 once the thermal-time requirement is met, and yield is biomass multiplied
 179 by the harvest index. Because SIMPLE uses a linear radiation-use efficiency,
 180 it does not credit the higher light-use efficiency of shaded canopies. Shading
 181 losses are therefore, if anything, overestimated.

182 2.3. SALS: stress-aware light sharing

183 SALS learns two functions (Algorithm 1):

- 184 • a *design network* $g = D_\theta(\mathbf{z})$, a multilayer perceptron with two hidden
 185 layers of 32 units, which maps site descriptors $\mathbf{z}$ to a ground coverage
 186 ratio in $[0.05, 0.60]$. The descriptors are the crop (one-hot), latitude,
 187 mean irradiance, growing-season maximum temperature, precipitation
 188 and vapour-pressure deficit, the frequency of heat-stress days and the
 189 irrigated fraction;
- 190 • a *control network* $u_t = \pi_\omega(\mathbf{x}_t)$, a multilayer perceptron with two hidden
 191 layers of 64 units and a sigmoid output, which chooses the daily light-
 192 sharing level of Eq. (1) from information available on day t : relative
 193 thermal time TT/T_{sum} , radiation interception f_{Solar} , relative soil water,
 194 the previous day’s ARID, the day’s forecast of maximum temperature
 195 relative to T_{heat} and T_{opt} , irradiance and ET_0 , the irrigation status, the
 196 design g , latitude, day of season and the crop.

197 Outside the cropping season the trackers backtrack for maximum electricity.
 198 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield
 199 retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional PV
 200 plant. SALS solves

$$\max_{\theta, \omega} \mathbb{E}_i[e_i] \quad \text{s.t.} \quad r_i \geq \rho \quad \forall i, \quad (3)$$

201 where ρ is the food-security floor ($\rho = 0.9$ unless stated otherwise). It does
 202 so with the penalty objective

$$\mathcal{L}(\theta, \omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu \max\{0, \rho - r_i\}^2 \right], \quad \mu = 30. \quad (4)$$

203 Because r_i and e_i are outputs of the differentiable simulator, the exact gradi-
 204 ent of Eq. (4) with respect to (θ, ω) is obtained by back-propagation through
 205 the 365 daily steps of every season (back-propagation through time). This
 206 is the key difference from reinforcement learning with black-box simulators
 207 (Gautron et al., 2022), which estimates gradients from sampled returns. Each
 208 minibatch contains 512 site-seasons, each duplicated into a rainfed and an
 209 irrigated field, drawn from one of the four climates in turn. The networks
 210 are trained with Adam (Kingma and Ba, 2015) (learning rate 3×10^{-3} , co-
 211 sine decay, gradient-norm clipping at 1) for 400 steps, i.e. about 50 passes

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalty μ

- 1: initialise design network D_θ and control network π_ω
 - 2: **for** step = $1, \dots, N$ **do**
 - 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated rows)
 - 4: $g_i \leftarrow D_\theta(\mathbf{z}_i)$; initialise crop and soil states
 - 5: **for** day $t = 1, \dots, 365$ **do**
 - 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it})$ if the crop is in the field, else 0
 - 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
 - 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
 - 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}$, $e_i \leftarrow E_i/E_{\text{PV},i}$; loss $\mathcal{L}$ (Eq. 4)
 - 10: back-propagate through the season; Adam update of (θ, ω)
-

212 over the training seasons. Growth-stage logistic functions are evaluated in a
213 numerically stable form, so that gradients remain finite even under extreme
214 heat.

215 2.4. Baselines

216 We compared SALS with the following references and baselines.

- 217 • *Open field*: no modules ($r = 1$, $e = 0$, LER = 1).
- 218 • *PV plant*: conventional ground-mounted PV without a crop ($e = 1$,
219 LER = 1).
- 220 • *AV static (rule)*: backtracking trackers with one GCR per crop, the
221 largest value on a grid $\{0.05, 0.10, \dots, 0.50\}$ whose mean retention on
222 the training seasons reaches ρ . This mimics a design guideline.
- 223 • *AV static (oracle GCR)*: backtracking trackers with the largest GCR
224 that meets ρ on average over each test site’s own seasons. The GCR is
225 chosen with knowledge of the test weather.
- 226 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
227 cropping season, with the per-site oracle GCR.

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a). T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °Cd, RUE in gMJ⁻¹; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

- *Stress rule*: an expert rule that shares light ($u = 1$) during the main growth phase ($0.1 < \text{TT}/T_{\text{sum}} < 0.9$) except on days with a heat-stress forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules track the sun and shade the crop. It uses the per-site oracle GCR.
- *Open-loop optimisation (perfect foresight)*: the GCR and all 365 daily controls of every test season are optimised directly by gradient descent through the simulator (150 Adam steps, learning rate 0.1) with perfect knowledge of that season’s weather. It is a per-season trajectory optimisation that no deployable controller could implement. Because the problem is non-convex and each season is optimised in isolation, it is not guaranteed to be an upper bound.

The three oracle-GCR baselines and the open-loop optimisation therefore have access to the test weather, whereas SALS chooses its design from climate descriptors alone.

2.5. Evaluation protocol and metrics

Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Seasons sown in 2001–2014 at training sites were used for training, and seasons sown in 2015–2019 at test sites for evaluation, so that the test set is out of sample in both space and time. We report the mean electricity (MWh ha⁻¹ yr⁻¹) and relative electricity e , the mean yield retention r , the *compliance* (share of test seasons with $r \geq \rho$), and the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011). Differences between methods were tested with paired Wilcoxon signed-rank tests over test seasons, and uncertainty is given as bootstrap 95% confidence intervals over sites.

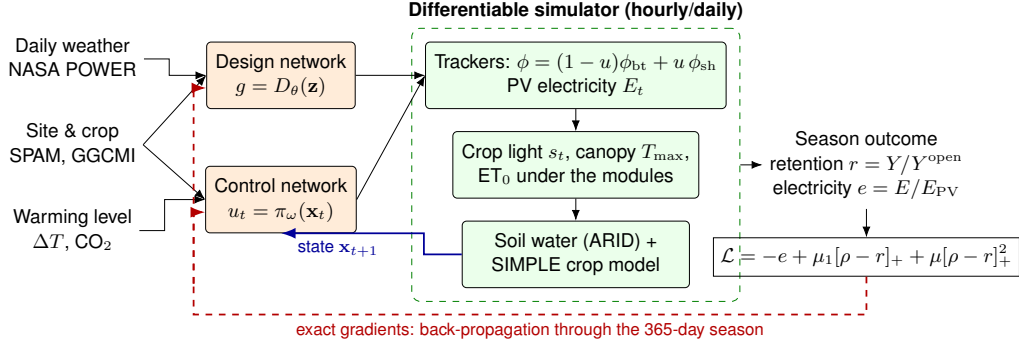


Figure 1: SALS. A design network chooses the ground coverage ratio g of single-axis tracker rows from site descriptors, and a control network chooses the daily light-sharing level u_t from the crop, soil and weather state. The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season to both networks.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 and Fig. 3 compare the methods on 461 test seasons: 2015–2019 at geographically held-out sites, excluding seasons in which the open-field crop failed ($< 0.2 \text{ t ha}^{-1}$). Under the baseline climate, the static agrivoltaic designs that keep 90% of the yield must use very sparse arrays (GCR ≈ 0.10). They produce $361\text{--}377 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, a quarter of the electricity of a conventional PV plant, and reach an LER of 1.18–1.20. Rotating the trackers to share light with the crop allows denser arrays. The seasonal-sharing and stress rules reach 629 and $707 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.37 and 1.42), even though their GCR is chosen with knowledge of the test weather. SALS chooses a GCR of 0.34 on average and produces $1019 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, 72% of a conventional PV plant, with a mean yield retention of 0.909. Its LER of 1.63 (95% CI 1.57–1.70) is 0.21 higher than that of the best expert rule and 0.43 higher than that of the static design ($p < 10^{-4}$, paired Wilcoxon tests, Table A.4). Per-season open-loop optimisation with perfect foresight reached an LER of 1.55. It remained below SALS because optimising 365 daily controls separately for each season is a harder, non-convex problem than learning one policy from thousands of seasons.

The gain comes with a trade-off that must be stated clearly. SALS meets

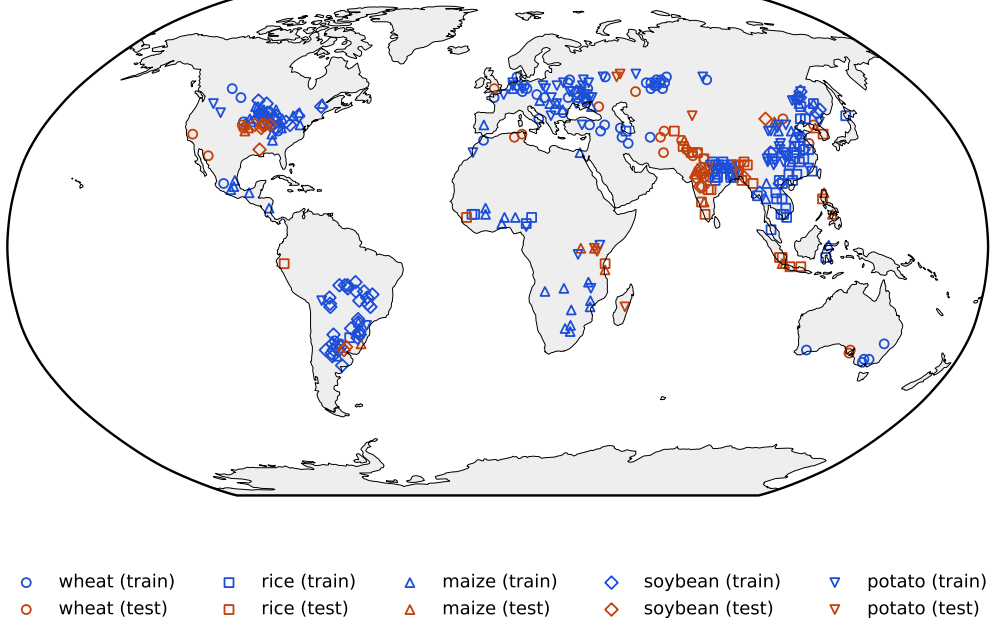


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites (10° blocks).

273 the floor *on average*: its mean retention is 0.90–0.91 in all four climates. It
 274 meets the floor in only 45–49% of individual seasons, however, whereas the
 275 rule-based baselines meet it in 78–85% of seasons. The penalty objective
 276 (Eq. 4) lets SALS go slightly below the floor in some seasons in exchange for
 277 much more electricity. The rules stay above the floor with a margin (mean
 278 $r \approx 0.92$) and produce 30–65% less electricity. The floor in Eq. (3) is a per-
 279 season constraint; a formulation that guarantees it in a given share of seasons,
 280 i.e. a chance constraint, would be the natural extension for deployments
 281 where regulation is enforced every season (Section 4.3).

282 3.2. What the policy learned

283 Fig. 4 shows the learned control for two of the hottest test seasons under
 284 +3 °C. For rainfed maize in the Nile Delta, SALS keeps the trackers in energy
 285 mode during crop establishment, opens them fully ($u \approx 1$) during the canopy-
 286 building phase, when intercepted light is most valuable, and returns to energy
 287 mode in the second half of the season, when maximum temperatures exceed

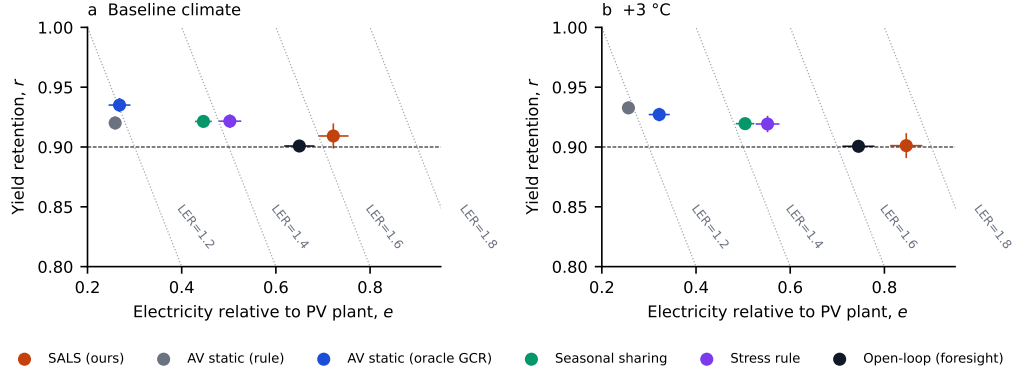


Figure 3: Electricity–food trade-off on held-out sites and years. Points are means with 95% confidence intervals; dotted lines are iso- LER lines; the dashed line is the 90% yield-retention floor.

288 40 °C and the shade protects the canopy. For irrigated wheat in north-western
 289 India, it shares light throughout the cool vegetative phase and shades the crop
 290 during the hot grain-filling period before harvest. Over all active crop days
 291 of the +3 °C test seasons, the mean light-sharing level is 0.33 on days with
 292 heat stress and 0.46 on other days. The policy has thus learned to trade light
 293 for cooling when heat, not light, limits growth. The design network chooses
 294 denser arrays for hotter and drier sites and for maize and soybean (mean
 295 GCR 0.42 and 0.39 at present climate) than for wheat and potato (0.32). In
 296 these crops, heat and drought limit yield more than light does.

297 3.3. Agrivoltaics under global warming

298 Warming changes the balance between light and heat. With the same
 299 trained networks, SALS increases its electricity from 1019 to 1079, 1106
 300 and 1173 MWh ha⁻¹ yr⁻¹ at +1.5, +2 and +3 °C (from 72% to 85% of a
 301 conventional PV plant) while keeping the mean retention relative to the
 302 same-climate open field at 0.90 (Table 2). Its LER rises from 1.63 to 1.75,
 303 whereas that of the static design stays at about 1.2. Two effects cause this.
 304 Warmer seasons are shorter, so the crop occupies the field for fewer days,
 305 and heat and drought stress make light less limiting, so more shade can be
 306 tolerated or is even beneficial. Across all 400 sites, the SALS system yields
 307 more than the open field in 13–14% of seasons, rising to 31–35% for maize
 308 and 25–28% for soybean, mostly at hot, rainfed sites (Fig. 6b).

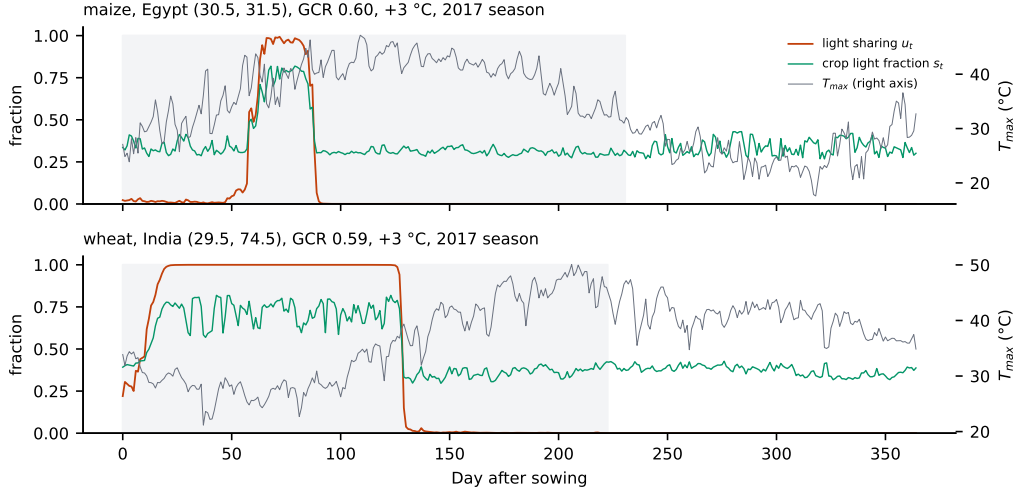


Figure 4: Learned daily control for two hot test seasons under +3 °C. Shading marks the cropping season. Orange: light-sharing level u_t chosen by SALS; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

309 Agrivoltaics is not, however, a remedy for the aggregate warming damage.
 310 Summed over sites, open-field production of maize and rice falls by 18 and
 311 17% at +3 °C in our simulations, and SALS production by 23 and 26% relative
 312 to the baseline open field, i.e. 94 and 90% of the same-climate open field
 313 (Fig. 5). Potato and wheat, which benefit from warmer conditions at cool
 314 sites and from CO₂ fertilisation in the model, gain in the open field, and
 315 SALS keeps 87–89% of this production. The benefit of SALS under warming
 316 is that it maintains the food-security floor while producing more electricity
 317 per hectare. It does not reverse the climate impact on yields.

318 3.4. Global electricity potential

319 Fig. 6 maps the electricity and LER of SALS at all 400 sites. Electricity
 320 ranges from a few hundred ha⁻¹ yr⁻¹ at high-latitude wheat sites to more than
 321 2000 MWh ha⁻¹ yr⁻¹ at sunny, hot maize and soybean sites in South Asia,
 322 Africa and the Americas, where the LER exceeds 2. Because the sites were
 323 sampled in proportion to harvested area, their mean electricity per hectare
 324 can be scaled to the global area of each crop. Deploying SALS agrivoltaics
 325 on 5% of the harvested area of the five crops (33 Mha) would generate about
 326 31,000 TWh yr⁻¹ at present climate and 37,000 TWh yr⁻¹ at +3 °C, while
 327 retaining on average 93% of the crop yield on that land. Static agrivoltaic

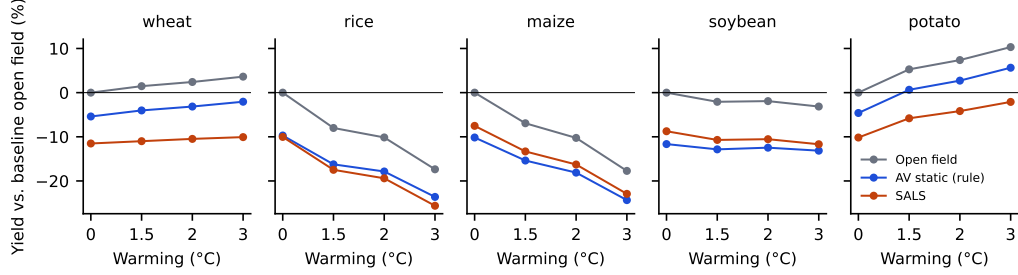


Figure 5: Production relative to the baseline open field (sum over sites, test years) as a function of warming, for the open field, static agrivoltaics and SALS.

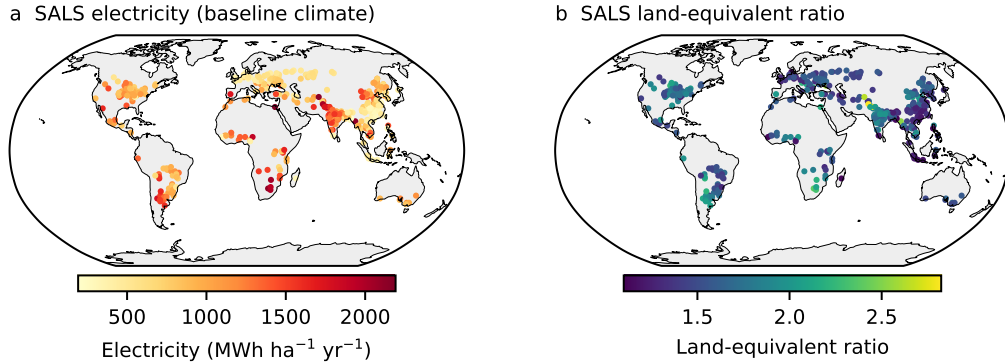


Figure 6: SALS electricity (a) and land-equivalent ratio (b) at the 400 sites (baseline climate, mean of seasons 2015–2019).

328 designs with the same yield floor would generate about $11,700 \text{ TWh yr}^{-1}$.
 329 For comparison, global electricity generation was about $30,000 \text{ TWh}$ in 2023
 330 (Ember, 2024). These are technical potentials: they ignore grid integration,
 331 storage, land tenure and economic constraints (Section 4.3). They show,
 332 however, that light sharing makes the food-compatible share of cropland
 333 large enough to matter for global energy security.

334 3.5. Sensitivity and ablation

335 The canopy-cooling coefficient κ is the most uncertain microclimate pa-
 336 rameter. Evaluated without retraining at $\kappa = 0, 1, 2$ and 3°C , the electricity
 337 of SALS barely changes ($1026\text{--}1022 \text{ MWh ha}^{-1} \text{yr}^{-1}$ at present climate). Its
 338 mean retention changes from 0.871 to 0.931 at present climate and from
 339 0.847 to 0.927 at $+3^\circ\text{C}$. Without any cooling benefit, the policy therefore

340 under-shoots the floor by 3–5 points. The heat-protection effect is thus an
 341 important part of the benefit under warming, and field measurements of
 342 canopy temperature under trackers are needed to pin it down.

343 Table 3 reports three ablations. Each variant was trained with the same
 344 budget of 200 steps as a reference SALS model and evaluated on the held-out
 345 seasons.

- 346 • *Without stress features*: the control network sees only phenology, time,
 347 design and crop, with soil water, ARID and the maximum-temperature,
 348 irradiance and ET_0 forecasts masked. At similar mean retention, this
 349 variant produced 28% less electricity (700 vs 971 MWh ha⁻¹ yr⁻¹) and
 350 lowered the LER from 1.59 to 1.40 at present climate and from 1.67 to
 351 1.42 at +3 °C. A stress-blind policy performs like the expert rules: it
 352 learns *when* in the season to share light, but it cannot respond to the
 353 heat and drought of a particular season.
- 354 • *Without the design network*: every site uses a GCR of 0.3, close to the
 355 mean GCR chosen by SALS, and only the controller is learned. The
 356 LER fell to 1.42 and 1.43, and compliance dropped to 31–38%. The
 357 benefit of SALS therefore comes from matching the array density to the
 358 site’s climate and crop: denser at hot, dry sites where shade is cheap,
 359 sparser where light limits growth. The same mean density everywhere
 360 does not achieve this.
- 361 • *Trained on the baseline climate only*: the networks were trained on the
 362 present climate alone and evaluated in all four climates. They did not
 363 learn to exploit the heat-protection value of shade. Their electricity
 364 stayed at about 670 MWh ha⁻¹ yr⁻¹ and their LER at 1.38–1.40 under
 365 all warming levels, whereas the multi-climate model’s LER rose from
 366 1.59 to 1.67. This variant was weaker even under the present climate.
 367 With a single training seed we cannot separate a regularising effect of
 368 the more diverse multi-climate data from optimisation variance, but
 369 the result shows that training on projected climates is needed for a
 370 policy that remains effective as the climate warms.

371 Training longer (400 instead of 200 steps) added a further 0.04–0.08 to
 372 the LER.

373 Finally, we retrained SALS (200 steps) with a softer floor of $\rho = 0.8$ and
 374 re-evaluated all baselines with the same floor. The baselines’ GCR grids and

375 rules were recalibrated for $\rho = 0.8$; the static rule then selects GCR 0.15–0.30.
 376 Under the present climate, the static design reached an LER of 1.35–1.36, sea-
 377 sonal sharing 1.59 and the stress rule 1.68 (1184 MWh ha⁻¹ yr⁻¹). SALS chose
 378 much denser arrays (mean GCR 0.55) and produced 1604 MWh ha⁻¹ yr⁻¹,
 379 15% more than a conventional PV plant with GCR 0.4, at a mean reten-
 380 tion of 0.815. Its LER was 1.97, rising to 2.03 at +3 °C. The ranking of the
 381 methods is therefore robust to the choice of the food-security floor. So is
 382 the main weakness of SALS: only 38–41% of its seasons met the 80% floor,
 383 against 74–85% for the rules. The floor thus steers where SALS operates
 384 on the energy–food frontier, but a penalised average does not guarantee it
 385 season by season.

386 4. Discussion

387 4.1. *Why learning the control matters*

388 The results show that most of the electricity that agrivoltaics can add
 389 without compromising food production comes from *when* light is shared
 390 rather than from the array geometry alone. With static trackers, the yield
 391 floor forces sparse arrays that cover a tenth of the ground. Rules that share
 392 light during the season double the electricity, and a learned, state-dependent
 393 schedule nearly triples it. SALS concentrates the crop’s light in the phases
 394 where intercepted radiation is converted into biomass most efficiently (canopy
 395 build-up) and when light, rather than heat or water, limits growth. In the
 396 remaining days it produces electricity and shades the crop against heat. This
 397 agrees with field observations that shading is less damaging, or even bene-
 398 ficial, when it coincides with heat or drought (Marrou et al., 2013; Barron-
 399 Gafford et al., 2019; Weselek et al., 2021), and with the dynamic-tracking
 400 experiments of Valle et al. (2017). SALS generalises this idea across crops and
 401 climates through a policy learned from data rather than from hand-crafted
 402 rules.

403 Differentiable simulation is what makes this learning tractable. A season
 404 has 365 coupled decisions and a single delayed outcome, the harvest. Exact
 405 gradients through the physics give credit to each daily decision without the
 406 variance of sampled returns that makes reinforcement learning with black-
 407 box crop simulators expensive (Gautron et al., 2022). Training one policy on
 408 about 4000 seasons in four climates took about 40 minutes on four CPU cores,
 409 and the policy transfers to unseen regions and years. Interestingly, the pol-
 410 icy outperformed per-season trajectory optimisation with perfect foresight.

411 Sharing one policy across thousands of seasons regularises the non-convex
412 control problem, a benefit of amortised optimisation.

413 *4.2. Implications for energy security and climate adaptation*

414 Agrivoltaics has been advocated as a way to decarbonise electricity with-
415 out taking land from food (Dinesh and Pearce, 2016; Adeh et al., 2019). Our
416 results quantify this potential under an explicit food-security floor and show
417 how strongly it depends on control. With static designs, 5% of the area of
418 five staple crops would supply about 40% of today’s global electricity. With
419 learned light sharing it would supply roughly all of it, on land that keeps
420 producing 93% of its food. The potential grows with warming because heat
421 makes shade less costly. Agrivoltaics therefore becomes more attractive as
422 the climate warms, in the regions of South Asia, Africa and the Americas
423 where both electricity demand and heat risks for crops are rising. SALS
424 does not, however, reverse the negative impact of warming on the aggregate
425 yields of maize and rice. Agrivoltaics complements breeding, irrigation and
426 management adaptation; it does not replace them.

427 *4.3. Limitations*

428 Several simplifications bound the conclusions. (i) The crop model is
429 generic. SIMPLE has a linear radiation-use efficiency, so it does not reward
430 the higher light-use efficiency of shaded canopies, and it represents heat stress
431 through maximum temperature only. Crop-specific shade responses (Marrou
432 et al., 2013; Weselek et al., 2021) and multi-model ensembles (Jägermeyr
433 et al., 2021) should be used to confirm the results. (ii) The microclimate
434 model is empirical: the canopy-cooling coefficient κ strongly affects yield
435 retention (Section 3.5) and needs field calibration for trackers. (iii) Warm-
436 ing was applied as uniform temperature shifts with fixed relative humidity,
437 precipitation and irradiance, not as bias-adjusted climate-model projections,
438 and cultivars and sowing dates were not adapted. (iv) Fields are either rain-
439 fed or fully irrigated, and irrigation water savings under the panels are not
440 credited. (v) SALS meets the yield floor on average but not in every season,
441 for both floors tested ($\rho = 0.9$ and 0.8). A chance-constrained or risk-averse
442 objective, e.g. a conditional value-at-risk of the retention, is needed where
443 the floor is enforced every year. (vi) The upscaled potentials are technical:
444 grid capacity, storage, the value of midday electricity, the cost of elevated
445 structures and farmers’ acceptance (Trommsdorff et al., 2021) will determine
446 the realisable share.

447 5. Conclusions

448 We have presented SALS, a learning algorithm that designs and controls
449 agrivoltaic systems end-to-end through a differentiable simulator of trackers,
450 PV output, crop light, microclimate, soil water and crop growth. Trained on
451 400 cropland sites of the world’s five major staple crops and four warming
452 levels, SALS produced 72–85% of the electricity of a conventional solar plant
453 while keeping the mean crop yield at 90% of the open field. This raised
454 the land-equivalent ratio from 1.2 for static designs and 1.4 for expert light-
455 sharing rules to 1.6–1.75 on geographically held-out sites and years. The
456 learned policy shares light during canopy development and shades the crop
457 during heat, and its advantage grows with warming. Scaled to 5% of the
458 harvested area of the five crops, learned agrivoltaics would have a technical
459 potential comparable to current global electricity generation. Future work
460 should add chance constraints for year-by-year food security, crop-specific
461 shade physiology and field validation of the learned control on commercial
462 tracker systems.

463 CRediT authorship contribution statement

464 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
465 analysis, Investigation, Data curation, Visualization, Writing – original draft,
466 Writing – review & editing.

467 Declaration of competing interest

468 The author has no known competing financial interests or personal re-
469 lationships that could have appeared to influence the work reported in this
470 paper.

471 Data and code availability

472 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
473 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
474 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
475 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
476 ments, together with the site list, is available from the author’s public reposi-
477 tory (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

478 **Funding**

479 This research did not receive any specific grant from funding agencies in
480 the public, commercial, or not-for-profit sectors.

481 **Appendix A. Statistical tests**

482 **References**

- 483 Adeh, E.H., Good, S.P., Calaf, M., Higgins, C.W., 2019. Solar pv power
484 potential is greatest over croplands. *Scientific Reports* 9, 11442. doi:10.
485 1038/s41598-019-47803-3.
- 486 Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. Crop evapotranspira-
487 tion: Guidelines for computing crop water requirements. *FAO Irrigation*
488 *and Drainage Paper* 56. Food and Agriculture Organization of the United
489 Nations. Rome.
- 490 Amaducci, S., Yin, X., Colauzzi, M., 2018. Agrivoltaic systems to optimise
491 land use for electric energy production. *Applied Energy* 220, 545–561.
492 doi:10.1016/j.apenergy.2018.03.081.
- 493 Barron-Gafford, G.A., Pavao-Zuckerman, M.A., Minor, R.L., Sutter, L.F.,
494 Barnett-Moreno, I., Blackett, D.T., Thompson, M., Dimond, K., Gerlak,
495 A.K., Nabhan, G.P., Macknick, J.E., 2019. Agrivoltaics provide mutual
496 benefits across the food–energy–water nexus in drylands. *Nature Sustain-*
497 *ability* 2, 848–855. doi:10.1038/s41893-019-0364-5.
- 498 Collares-Pereira, M., Rabl, A., 1979. The average distribution of solar
499 radiation-correlations between diffuse and hemispherical and between daily
500 and hourly insolation values. *Solar Energy* 22, 155–164. doi:10.1016/
501 0038-092x(79)90100-2.
- 502 DIN, 2021. DIN SPEC 91434:2021-05: Agri-photovoltaic systems – require-
503 ments for primary agricultural use. Beuth Verlag, Berlin.
- 504 Dinesh, H., Pearce, J.M., 2016. The potential of agrivoltaic systems. *Renew-*
505 *able and Sustainable Energy Reviews* 54, 299–308. doi:10.1016/j.rser.
506 2015.10.024.

507 Dupraz, C., Marrou, H., Talbot, G., Dufour, L., Nogier, A., Ferard, Y., 2011.
508 Combining solar photovoltaic panels and food crops for optimising land
509 use: Towards new agrivoltaic schemes. *Renewable Energy* 36, 2725–2732.
510 doi:10.1016/j.renene.2011.03.005.

511 Elamri, Y., Cheviron, B., Lopez, J.M., Dejean, C., Belaud, G., 2018. Water
512 budget and crop modelling for agrivoltaic systems: Application to irrigated
513 lettuces. *Agricultural Water Management* 208, 440–453. doi:10.1016/j.
514 agwat.2018.07.001.

515 Ember, 2024. Global Electricity Review 2024. Technical Report. Ember.
516 London.

517 Gautron, R., Maillard, O.A., Preux, P., Corbeels, M., Sabbadin, R., 2022.
518 Reinforcement learning for crop management support: Review, prospects
519 and challenges. *Computers and Electronics in Agriculture* 200, 107182.
520 doi:10.1016/j.compag.2022.107182.

521 Hassanpour Adeg, E., Selker, J.S., Higgins, C.W., 2018. Remarkable agri-
522 voltaic influence on soil moisture, micrometeorology and water-use effi-
523 ciency. *PLOS ONE* 13, e0203256. doi:10.1371/journal.pone.0203256.

524 Jägermeyr, J., Müller, C., Ruane, A.C., Elliott, J., Balkovic, J., Castillo, O.,
525 Faye, B., Foster, I., Folberth, C., Franke, J.A., Fuchs, K., Guarin, J.R.,
526 Heinke, J., Hoogenboom, G., Iizumi, T., Jain, A.K., Kelly, D., Khabarov,
527 N., Lange, S., Lin, T.S., Liu, W., Mialyk, O., Minoli, S., Moyer, E.J.,
528 Okada, M., Phillips, M., Porter, C., Rabin, S.S., Scheer, C., Schneider,
529 J.M., Schyns, J.F., Skalsky, R., Smerald, A., Stella, T., Stephens, H.,
530 Webber, H., Zabel, F., Rosenzweig, C., 2021. Climate impacts on global
531 agriculture emerge earlier in new generation of climate and crop models.
532 *Nature Food* 2, 873–885. doi:10.1038/s43016-021-00400-y.

533 Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang,
534 L., 2021. Physics-informed machine learning. *Nature Reviews Physics* 3,
535 422–440. doi:10.1038/s42254-021-00314-5.

536 Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization,
537 in: International Conference on Learning Representations.

538 Liu, B.Y., Jordan, R.C., 1960. The interrelationship and characteristic dis-
539 tribution of direct, diffuse and total solar radiation. *Solar Energy* 4, 1–19.
540 doi:10.1016/0038-092x(60)90062-1.

541 Lorenzo, E., Narvarte, L., Muñoz, J., 2011. Tracking and back-tracking.
542 Progress in Photovoltaics: Research and Applications 19, 747–753. doi:10.
543 1002/pip.1085.

544 Marrou, H., Guillioni, L., Dufour, L., Dupraz, C., Wery, J., 2013. Microcli-
545 mate under agrivoltaic systems: Is crop growth rate affected in the partial
546 shade of solar panels? *Agricultural and Forest Meteorology* 177, 117–132.
547 doi:10.1016/j.agrformet.2013.04.012.

548 Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen,
549 T., Lin, Z., Gimelshein, N., Antiga, L., et al., 2019. PyTorch: An impera-
550 tive style, high-performance deep learning library, in: *Advances in Neural*
551 *Information Processing Systems*, pp. 8024–8035.

552 Sparks, A., 2018. nasapower: A nasa power global meteorology, surface solar
553 energy and climatology data client for r. *Journal of Open Source Software*
554 3, 1035. doi:10.21105/joss.01035.

555 Trommsdorff, M., Kang, J., Reise, C., Schindele, S., Bopp, G., Ehmann, A.,
556 Weselek, A., Högy, P., Obergfell, T., 2021. Combining food and energy
557 production: Design of an agrivoltaic system applied in arable and vegetable
558 farming in germany. *Renewable and Sustainable Energy Reviews* 140,
559 110694. doi:10.1016/j.rser.2020.110694.

560 Valle, B., Simonneau, T., Sourd, F., Pechier, P., Hamard, P., Frisson, T.,
561 Ryckewaert, M., Christophe, A., 2017. Increasing the total productivity of
562 a land by combining mobile photovoltaic panels and food crops. *Applied*
563 *Energy* 206, 1495–1507. doi:10.1016/j.apenergy.2017.09.113.

564 Weselek, A., Bauerle, A., Hartung, J., Zikeli, S., Lewandowski, I., Högy, P.,
565 2021. Agrivoltaic system impacts on microclimate and yield of different
566 crops within an organic crop rotation in a temperate climate. *Agronomy*
567 *for Sustainable Development* 41, 59. doi:10.1007/s13593-021-00714-y.

568 Weselek, A., Ehmann, A., Zikeli, S., Lewandowski, I., Schindele, S., Högy,
569 P., 2019. Agrophotovoltaic systems: applications, challenges, and op-

570 opportunities. a review. *Agronomy for Sustainable Development* 39, 35.
571 doi:10.1007/s13593-019-0581-3.

572 Woli, P., Jones, J.W., Ingram, K.T., Fraisse, C.W., 2012. Agricultural
573 reference index for drought (arid). *Agronomy Journal* 104, 287–300.
574 doi:10.2134/agronj2011.0286.

575 Yu, Q., You, L., Wood-Sichra, U., Ru, Y., Joglekar, A.K.B., Fritz, S., Xiong,
576 W., Lu, M., Wu, W., Yang, P., 2020. A cultivated planet in 2010 – part 2:
577 The global gridded agricultural-production maps. *Earth System Science*
578 *Data* 12, 3545–3572. doi:10.5194/essd-12-3545-2020.

579 Zhang, L., Geng, X., Ding, H., Cao, K., Wang, L., Chen, H., Deng, L., Wu,
580 C., Xiao, M., Bao, E., 2026. Multi-objective optimization of photovoltaic
581 array layouts on farmland via a hybrid algorithm framework for enhanced
582 agricultural production and energy generation. *Artificial Intelligence in*
583 *Agriculture* doi:10.1016/j.aiaa.2026.09.002.

584 Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D.B., Huang, Y., Huang, M.,
585 Yao, Y., Bassu, S., Ciais, P., Durand, J.L., Elliott, J., Ewert, F., Janssens,
586 I.A., Li, T., Lin, E., Liu, Q., Martre, P., Müller, C., Peng, S., Peñuelas,
587 J., Ruane, A.C., Wallach, D., Wang, T., Wu, D., Liu, Z., Zhu, Y., Zhu,
588 Z., Asseng, S., 2017. Temperature increase reduces global yields of major
589 crops in four independent estimates. *Proceedings of the National Academy*
590 *of Sciences* 114, 9326–9331. doi:10.1073/pnas.1701762114.

591 Zhao, C., Liu, B., Xiao, L., Hoogenboom, G., Boote, K.J., Kassie, B.T.,
592 Pavan, W., Shelia, V., Kim, K.S., Hernandez-Ochoa, I.M., Wallach, D.,
593 Porter, C.H., Stockle, C.O., Zhu, Y., Asseng, S., 2019. A simple crop
594 model. *European Journal of Agronomy* 104, 97–106. doi:10.1016/j.eja.
595 2019.01.009.

Table 2: Held-out performance (test sites, seasons 2015–2019). GCR: mean ground coverage ratio; E : electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; compliance: share of seasons with $r \geq 0.9$; LER = $r + e$ with bootstrap 95% CI over sites. Oracle-GCR baselines choose the GCR with knowledge of the test weather; the open-loop optimisation also knows the daily weather of each test season.

Method	GCR	E	e	r	Compliance (%)	LER [95% CI]
<i>Baseline</i>						
AV static (rule)	0.10	361	0.26	0.920	60.3	1.18 [1.16, 1.20]
AV static (oracle GCR)	0.10	377	0.27	0.935	85.2	1.20 [1.16, 1.25]
Seasonal sharing	0.26	629	0.45	0.921	82.2	1.37 [1.33, 1.41]
Stress rule	0.23	707	0.50	0.922	80.9	1.42 [1.37, 1.48]
SALS (ours)	0.34	1019	0.72	0.909	49.5	1.63 [1.57, 1.70]
Open-loop (foresight)	0.30	916	0.65	0.901	28.6	1.55 [1.49, 1.61]
<i>+1.5 °C</i>						
AV static (rule)	0.10	355	0.26	0.924	66.7	1.18 [1.16, 1.20]
AV static (oracle GCR)	0.11	385	0.28	0.931	83.5	1.21 [1.17, 1.25]
Seasonal sharing	0.28	647	0.47	0.918	79.2	1.38 [1.34, 1.43]
Stress rule	0.23	722	0.52	0.919	77.7	1.44 [1.39, 1.49]
SALS (ours)	0.36	1079	0.77	0.901	46.8	1.67 [1.61, 1.74]
<i>+2 °C</i>						
AV static (rule)	0.10	354	0.26	0.926	69.1	1.18 [1.16, 1.20]
AV static (oracle GCR)	0.11	393	0.29	0.929	83.5	1.22 [1.17, 1.26]
Seasonal sharing	0.28	659	0.48	0.917	78.9	1.39 [1.35, 1.44]
Stress rule	0.23	728	0.53	0.918	77.2	1.44 [1.39, 1.50]
SALS (ours)	0.37	1106	0.79	0.898	46.3	1.69 [1.62, 1.76]
<i>+3 °C</i>						
AV static (rule)	0.10	353	0.26	0.933	74.2	1.19 [1.17, 1.21]
AV static (oracle GCR)	0.12	442	0.32	0.927	80.1	1.25 [1.20, 1.30]
Seasonal sharing	0.30	694	0.50	0.920	78.1	1.42 [1.38, 1.47]
Stress rule	0.24	762	0.55	0.919	77.7	1.47 [1.42, 1.53]
SALS (ours)	0.39	1173	0.85	0.901	45.1	1.75 [1.68, 1.82]
Open-loop (foresight)	0.34	1028	0.74	0.901	25.8	1.65 [1.58, 1.71]

Table 3: Ablation on held-out seasons. E : electricity (MWh ha⁻¹ yr⁻¹); r : mean yield retention; C: compliance (% of seasons with $r \geq 0.9$). All ablations were trained for 200 steps and should be compared with SALS (200 steps).

Variant	Baseline climate				+3 °C			
	E	r	C (%)	LER	E	r	C (%)	LER
SALS, full training (400 steps)	1019	0.909	49.5	1.63	1173	0.901	45.1	1.75
SALS (200 steps)	971	0.897	47.3	1.59	1079	0.892	43.5	1.67
w/o stress features	700	0.898	33.4	1.40	706	0.906	42.7	1.42
w/o design network (GCR 0.3)	734	0.889	31.2	1.42	719	0.900	38.1	1.43
trained on baseline climate only	668	0.904	41.4	1.38	676	0.910	46.6	1.40

Table A.4: Paired differences between SALS and each baseline on the held-out test seasons (SALS minus baseline) and two-sided Wilcoxon signed-rank p -values.

Climate	Baseline	ΔLER	p	Δe	p	n
Baseline	AV static (rule)	+0.452	$< 10^{-4}$	+0.463	$< 10^{-4}$	461
Baseline	AV static (oracle GCR)	+0.428	$< 10^{-4}$	+0.454	$< 10^{-4}$	461
Baseline	Seasonal sharing	+0.264	$< 10^{-4}$	+0.276	$< 10^{-4}$	461
Baseline	Stress rule	+0.207	$< 10^{-4}$	+0.220	$< 10^{-4}$	461
Baseline	Open-loop (foresight)	+0.081	$< 10^{-4}$	+0.072	0.0012	461
+1.5 °C	AV static (rule)	+0.493	$< 10^{-4}$	+0.515	$< 10^{-4}$	462
+1.5 °C	AV static (oracle GCR)	+0.463	$< 10^{-4}$	+0.493	$< 10^{-4}$	462
+1.5 °C	Seasonal sharing	+0.290	$< 10^{-4}$	+0.306	$< 10^{-4}$	462
+1.5 °C	Stress rule	+0.235	$< 10^{-4}$	+0.252	$< 10^{-4}$	462
+2 °C	AV static (rule)	+0.510	$< 10^{-4}$	+0.537	$< 10^{-4}$	460
+2 °C	AV static (oracle GCR)	+0.477	$< 10^{-4}$	+0.508	$< 10^{-4}$	460
+2 °C	Seasonal sharing	+0.299	$< 10^{-4}$	+0.318	$< 10^{-4}$	460
+2 °C	Stress rule	+0.249	$< 10^{-4}$	+0.269	$< 10^{-4}$	460
+3 °C	AV static (rule)	+0.558	$< 10^{-4}$	+0.590	$< 10^{-4}$	457
+3 °C	AV static (oracle GCR)	+0.498	$< 10^{-4}$	+0.524	$< 10^{-4}$	457
+3 °C	Seasonal sharing	+0.324	$< 10^{-4}$	+0.342	$< 10^{-4}$	457
+3 °C	Stress rule	+0.276	$< 10^{-4}$	+0.295	$< 10^{-4}$	457
+3 °C	Open-loop (foresight)	+0.102	$< 10^{-4}$	+0.101	$< 10^{-4}$	457